



Food Ordering Behaviour and Trends Analysis Report

A data analytics project report

Certificate & Acknowledgement

This report documents the Food Ordering Behaviour and Trends Analysis project. We hereby acknowledge the contributions of all individuals and resources that supported this endeavor.

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Team Members

This project was a collaborative effort by the following team members:

- **[Team Member 1 Syed Wahid Aslam]:** Project Lead, Data Analysis
- **[Team Member 2 Shaik Athif Basha]:** Data Visualization, Dashboard Design
- **[Team Member 3 Gadimurthi Narendra Syed Vahid Aslam]:** Data Preprocessing, Statistical Analysis
- **[Team Member 4 Syed Mohammed Zaheer]:** Report Writing, Presentation
- **[Team Member 5 Thunga Manmanth Reddy]:** Documentation

Project Overview

The Food Ordering Behaviour and Trends Analysis project aims to delve into the intricate patterns of consumer behavior within the online food ordering ecosystem. By leveraging data analytics techniques, this project seeks to uncover key trends, understand customer preferences, and identify

factors influencing ordering decisions. The ultimate goal is to provide actionable insights that can assist food businesses in optimizing their strategies, enhancing customer satisfaction, and driving growth in the competitive food delivery market.

Problem Understanding

The rapidly evolving landscape of the food industry, coupled with the proliferation of online food ordering platforms, presents both opportunities and challenges for businesses. Understanding the nuances of customer ordering behavior is paramount for survival and success. Key challenges include:

- Identifying popular cuisines and dishes.
- Understanding peak ordering times and days.
- Analyzing the impact of promotions and discounts.
- Segmenting customers based on their ordering habits.
- Predicting future demand and trends.

This project addresses these challenges by providing a data-driven approach to comprehending these dynamics.

Dataset Description

The dataset used for this analysis comprises detailed records of food orders placed through an online platform. It includes the following key attributes:

- **Order ID:** Unique identifier for each order.
- **Customer ID:** Unique identifier for each customer.
- **Restaurant ID:** Unique identifier for each restaurant.
- **Order Date & Time:** Timestamp of when the order was placed.
- **Delivery Location:** Geographic coordinates or area of delivery.
- **Cuisine Type:** Category of cuisine ordered (e.g., Italian, Chinese, Indian).
- **Dish Name:** Specific name of the dish ordered.
- **Quantity:** Number of units of a specific dish.
- **Price:** Price of the dish.
- **Total Amount:** Total value of the order.
- **Payment Method:** Mode of payment used.
- **Customer Ratings:** Rating provided by the customer for the order/restaurant.

The dataset spans a period of [Specify Time Period, e.g., one year] and contains approximately [Specify Number] records, offering a comprehensive view of ordering patterns.

Technical Architecture

The technical architecture for this project is designed to facilitate efficient data processing, analysis, and visualization.

1. **Data Ingestion:** Raw data is extracted from the source database.
2. **Data Preprocessing:** Using Python libraries such as Pandas and NumPy, the data is cleaned, transformed, and prepared for analysis. This involves handling missing values, data type conversions, and feature engineering.
3. **Data Analysis:** Statistical methods and algorithms are employed using Python (Scikit-learn, Statsmodels) to identify trends, correlations, and patterns.
4. **Data Visualization:** Tableau is utilized as the primary tool for creating interactive dashboards and stories to represent the analyzed data in a visually intuitive manner.
5. **Deployment:** The Tableau dashboards and stories are published to Tableau Server/Public for accessibility and sharing.

Tableau Dashboard Analysis

A series of interactive dashboards have been developed in Tableau to provide a comprehensive overview of food ordering behavior. These dashboards visualize key metrics and trends, allowing users to explore the data dynamically.

- **Overall Trends Dashboard:** Displays aggregated data on total orders, revenue, and average order value over time. Includes filters for date range and cuisine type.
- **Customer Behavior Dashboard:** Focuses on customer segmentation, repeat order rates, and popular ordering times. Visualizes customer demographics and their ordering preferences.
- **Restaurant Performance Dashboard:** Analyzes the performance of different restaurants, highlighting top-performing establishments, popular dishes, and customer ratings.
- **Geographic Analysis Dashboard:** Maps order distribution, identifying areas with high demand and potential for expansion.

Each dashboard is designed with user-friendly navigation and clear visualizations to facilitate easy understanding of complex data.

Tableau Story Analysis

Complementing the dashboards, a Tableau Story has been crafted to guide stakeholders through a narrative of the project's findings. This story links the dashboards and individual visualizations to present a cohesive and compelling analysis.

The story flows through the following key themes:

1. **Introduction to Food Ordering Trends:** A high-level overview of the market and key growth drivers.
2. **Understanding the Customer:** Deep dive into customer demographics, preferences, and loyalty.
3. **Restaurant Dynamics:** Analysis of factors contributing to restaurant success.

4. **Geographic Hotspots:** Identification of key areas for growth and targeted marketing.
5. **Actionable Recommendations:** Synthesis of insights into strategic advice for businesses.

The story format ensures that the insights are presented in a logical sequence, making it easier for decision-makers to grasp the implications and potential actions.

Dashboard Insights

The analysis derived from the Tableau dashboards and story has yielded several critical insights:

- **Peak Ordering Times:** Orders are predominantly placed during lunch (12 PM - 2 PM) and dinner (6 PM - 8 PM) hours, with a significant surge on weekends.
- **Popular Cuisines:** [List top 3-5 popular cuisines, e.g., Indian, Italian, Chinese] consistently emerge as the most frequently ordered cuisines.
- **Impact of Promotions:** Discounted offers and promotional campaigns show a direct correlation with an increase in order volume, especially among price-sensitive customer segments.
- **Customer Loyalty:** A notable percentage of revenue is generated by repeat customers, emphasizing the importance of customer retention strategies.
- **Geographic Concentration:** Urban centers and densely populated areas exhibit the highest order density, indicating strong market penetration.
- **Rating Influence:** Higher customer ratings are strongly associated with increased order frequency and positive word-of-mouth referrals.

Skills & Tools

This project leveraged a combination of technical skills and industry-standard tools:

- **Programming Languages:** Python (for data manipulation and analysis)
- **Data Analysis Libraries:** Pandas, NumPy, Scikit-learn, Statsmodels
- **Data Visualization Tools:** Tableau (for dashboard and story creation)
- **Database Management:** [Mention any specific databases used, e.g., SQL]
- **Statistical Analysis:** Hypothesis testing, regression analysis, descriptive statistics.
- **Project Management:** Agile methodologies, collaborative tools (e.g., Trello, Jira - if applicable).

Project Statistics

- **Total Orders Analyzed:** [Number]
- **Number of Unique Customers:** [Number]
- **Number of Unique Restaurants:** [Number]
- **Average Order Value:** [Currency Symbol] [Value]
- **Customer Retention Rate:** [Percentage]
- **Peak Ordering Day:** [Day of the Week]
- **Most Popular Cuisine:** [Cuisine Name]

- **Data Processing Time:** [Time taken for preprocessing]
- **Dashboard Development Time:** [Time taken for dashboard development]

Conclusion

The Food Ordering Behaviour and Trends Analysis project has successfully provided a comprehensive understanding of consumer patterns within the online food delivery market. The insights derived from the data have highlighted key drivers of customer behavior, restaurant performance, and market dynamics. By leveraging advanced analytics and interactive visualizations, this project offers actionable recommendations for businesses seeking to optimize their operations, enhance customer engagement, and achieve sustainable growth in this competitive industry. The methodologies and tools employed herein serve as a robust framework for ongoing analysis and strategic decision-making.